

AURANGABAD, India

WMO: 430140

Lat: 19.863N

Lon: 75.398E

Elev: 1911

StdP: 13.71

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	51.9	54.4	35.6	32.7	76.8	38.8	37.2	75.2	12.3	77.8	10.0	78.2	0.1	340	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	22.5	104.6	73.0	103.0	72.9	101.2	72.5	80.5	96.3	78.7	92.3	77.5	89.4	6.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.7	149.4	86.8	75.5	143.5	84.0	74.7	139.5	82.4	45.9	96.3	43.9	92.2	42.7	89.7	88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.5	12.6	11.1	47.7	107.9	2.5	1.3	45.9	108.9	44.5	109.7	43.1	110.4	41.3	111.4		
(5)			43.2	83.1	2.6	2.6	41.4	85.0	39.9	86.6	38.5	88.0	36.6	89.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.2	70.4	75.4	82.5	88.7	90.6	84.2	79.1	77.9	78.7	78.1	73.9	70.3		
	DBStd	7.08	3.65	4.01	4.35	3.75	3.28	4.41	2.69	2.28	2.10	2.80	3.24	3.61		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	8	5	0	0	0	0	0	0	0	0	0	0	3		
	CDD50	10641	631	711	1006	1162	1258	1026	902	866	862	871	717	629		
	CDD65	5177	171	291	542	712	793	576	437	401	412	406	268	168		
	CDH74	59538	1795	3193	7017	10909	12442	6816	3353	2685	3182	3826	2561	1759		
	CDH80	30189	530	1496	4007	6989	8160	3402	963	649	981	1614	917	481		
(14) Wind	WSAvg	4.0	1.9	2.6	2.9	3.8	6.4	7.2	6.7	6.4	4.0	2.4	2.0	1.7		
(15) Precipitation	PrecAvg	35.8	0.1	0.1	0.4	0.3	1.0	6.5	7.9	8.3	6.9	3.3	0.7	0.4		
	PrecMax	55.1	0.8	0.7	3.8	1.4	8.7	19.6	18.0	15.7	13.4	15.1	4.3	4.0		
	PrecMin	23.3	0.0	0.0	0.0	0.0	0.0	1.7	1.6	2.6	2.6	0.3	0.0	0.0		
	PrecStd	7.8	0.2	0.2	0.8	0.4	1.9	3.7	4.1	3.3	2.9	3.1	1.3	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.6	95.4	102.1	106.2	107.2	104.1	92.4	89.7	91.3	93.6	90.1	88.8		
		MCWB	67.6	68.6	70.3	71.8	74.0	75.9	75.3	75.7	75.8	71.7	70.1	69.7		
	2%	DB	87.1	93.4	99.8	104.2	105.4	100.6	89.7	87.4	89.1	91.6	88.5	86.5		
		MCWB	66.7	67.8	69.5	72.9	73.2	75.7	75.4	75.1	75.2	71.5	69.8	67.9		
	5%	DB	85.1	91.1	97.8	102.5	103.7	97.2	87.5	85.6	87.3	89.8	87.0	84.7		
		MCWB	66.4	67.6	69.1	72.3	72.8	75.4	75.3	74.9	74.7	70.9	69.5	66.8		
	10%	DB	83.1	88.7	95.5	100.8	102.0	94.0	85.3	83.8	85.5	88.1	85.4	82.8		
		MCWB	65.5	67.7	68.5	71.6	73.0	75.5	75.1	74.6	74.7	70.6	69.0	65.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.8	75.8	78.2	83.2	83.7	81.3	78.3	78.2	79.2	77.6	75.9	73.6		
		MCDB	84.5	89.8	95.2	99.9	101.0	96.1	85.6	85.7	87.9	86.6	84.6	85.5		
	2%	WB	70.3	73.1	75.6	79.6	80.7	79.2	77.3	77.0	77.4	76.3	74.3	71.1		
		MCDB	83.1	87.8	93.7	99.0	97.3	91.6	84.8	83.9	85.1	84.8	83.3	82.3		
	5%	WB	68.4	71.1	73.3	77.0	78.6	77.9	76.4	76.0	76.5	75.3	73.0	69.3		
		MCDB	81.8	86.5	92.0	97.2	95.2	89.6	83.7	82.6	83.8	83.3	82.1	80.8		
	10%	WB	66.7	69.1	71.1	74.7	76.9	77.0	75.6	75.2	75.7	74.3	71.7	67.6		
		MCDB	80.6	84.8	90.6	95.2	93.4	88.3	82.5	81.6	82.5	81.9	80.7	79.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.9	26.2	25.2	23.8	22.5	16.6	11.2	11.0	13.3	20.1	23.9	26.2		
		MCDBR	27.7	28.4	26.4	24.4	23.3	21.6	15.3	14.1	16.5	23.1	25.8	27.5		
	5% WB	MCWBR	13.5	12.5	11.1	10.2	8.0	6.1	4.8	5.2	6.0	8.6	11.7	13.3		
		MCWBR	24.9	25.3	25.2	23.6	21.8	17.7	12.5	12.1	13.6	15.7	19.8	24.0		
(40) Clear-Sky Solar Irradiance	taub		0.443	0.453	0.490	0.530	0.542	0.584	0.562	0.536	0.523	0.520	0.493	0.449		
		taud	2.079	2.068	1.987	1.933	1.943	1.901	1.965	2.044	2.063	1.996	2.008	2.073		
	Ebn at Noon		261	267	261	252	246	234	239	247	249	244	244	254		
		Edh at Noon	48	51	58	62	61	63	59	55	53	55	51	47		
(44) All-Sky Solar Radiation	RadAvg		1513	1789	2057	2227	2264	1696	1268	1262	1508	1643	1508	1450		
	RadStd		48	61	86	54	75	156	120	93	107	85	90	47		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.81	+1.89	N/A	+0.94	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+0.81	+1.89	N/A	+0.94	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page